

Convert decimals to percentages

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Write 0.625 as a percentage. Copy or complete the first step.

2. Write 35% as a fraction in simplest form.

3. Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.

4. Miro says: Miro moves the decimal point without considering the value. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Write 0.625 as a percentage. Copy or complete the first step.

Answer 62.5%.

2. Write 35% as a fraction in simplest form.

Answer $\frac{7}{20}$.

3. Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.

Answer 0.58, $\frac{3}{5}$, 62%.

4. Miro says: Miro moves the decimal point without considering the value. Is this correct? Explain.

Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.

Convert decimals to percentages

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Write 0.625 as a percentage.

6. Check this result using an inverse, estimate or known fact: Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.

2. Write 35% as a fraction in simplest form.

7. Prove or explain your reasoning: Complete: $\frac{9}{25} = \frac{\quad}{\quad} = \frac{\quad}{\quad}\%$.

3. Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.

8. Identify and correct this misconception: Miro moves the decimal point without considering the value.

4. Solve this independently and show every stage: Write 0.625 as a percentage.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Write 35% as a fraction in simplest form.

10. Write and solve a similar question to: Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.

Teacher answers and checking prompts

1. Write 0.625 as a percentage.
Answer 62.5%.
2. Write 35% as a fraction in simplest form.
Answer $\frac{7}{20}$.
3. Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.
Answer 0.58, $\frac{3}{5}$, 62%.
4. Solve this independently and show every stage: Write 0.625 as a percentage.
Answer 62.5%.
5. Use a second method or representation for: Write 35% as a fraction in simplest form.
Answer Expected result: $\frac{7}{20}$. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.
Answer Expected result: 0.58, $\frac{3}{5}$, 62%. A valid check must be shown.
7. Prove or explain your reasoning: Complete: $\frac{9}{25} = \frac{\quad}{\quad} = \frac{\quad}{\quad}\%$.
Answer 0.36 and 36%. The explanation must show why the method works.
8. Identify and correct this misconception: Miro moves the decimal point without considering the value.
Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Order $\frac{3}{5}$, 0.58 and 62% from least to greatest.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Convert decimals to percentages

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$.

6. Create a new example that tests the same idea as: Order $3/5$, 0.58 and 62% from least to greatest.

2. Prove the answer to this question, rather than only calculating it: Write 0.625 as a percentage.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Write 35% as a fraction in simplest form.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 0.58 , $3/5$, 62% .

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro moves the decimal point without considering the value.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Complete: $9/25 = \underline{\quad} = \underline{\quad}\%$.
Answer 0.36 and 36%. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Write 0.625 as a percentage.
Answer Expected result: 62.5%. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Write 35% as a fraction in simplest form.
Answer Expected result: $7/20$. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 0.58, $3/5$, 62%.
Answer One valid reconstruction is: Order $3/5$, 0.58 and 62% from least to greatest. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro moves the decimal point without considering the value.
Answer Use a hundred-square model or multiplication and division by 100, then check against a known benchmark.
6. Create a new example that tests the same idea as: Order $3/5$, 0.58 and 62% from least to greatest.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.